



Swathi Sri R 250701769-2025-CSE ▾

S2**Started on** Friday, 11 September 2026, 4:15 PM**State** Finished**Completed on** Friday, 11 September 2026, 4:16 PM**Time taken** 36 secs**Marks** 1.00/1.00**Grade** 30.00 out of 30.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```

1 #include <stdio.h>
2 #include <stdlib.h>
3
4 int compare(const void *a, const void *b) {
5     return (*(int *)a - *(int *)b);
6 }
7
8 void solve_test_case() {
9     int n1, n2;
10    if (scanf("%d", &n1) != 1) return;
11    int *arr1 = (int *)malloc(n1 * sizeof(int));
12    for (int i = 0; i < n1; i++) {
13        if (scanf("%d", &arr1[i]) != 1) return;
14    }
15
16    if (scanf("%d", &n2) != 1) return;
17    int *arr2 = (int *)malloc(n2 * sizeof(int));
18    for (int j = 0; j < n2; j++) {
19        if (scanf("%d", &arr2[j]) != 1) return;
20    }
21
22    qsort(arr1, n1, sizeof(int), compare);

```

```

22  qsort(arr1, n1, sizeof(int), compare);
23  qsort(arr2, n2, sizeof(int), compare);
24
25  int *result = (int *)malloc((n1 < n2 ? n1 : n2) * sizeof(int));
26  int k = 0;
27
28  int i = 0, j = 0;
29  while (i < n1 && j < n2) {
30      if (arr1[i] == arr2[j]) {
31          if (k == 0 || result[k - 1] != arr1[i]) {
32              result[k++] = arr1[i];
33          }
34          i++;
35          j++;
36      } else if (arr1[i] < arr2[j]) {
37          i++;
38      } else {
39          j++;
40      }
41  }
42
43  for (int x = 0; x < k; x++) {
44      printf("%d", result[x]);
45      if (x < k - 1) {
46          printf(" ");
47      }
48  }
49  printf("\n");
50
51  free(arr1);
52  free(arr2);

```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.



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